Coordination in State-Dependent Distributed Networks: The Two-Agent Case

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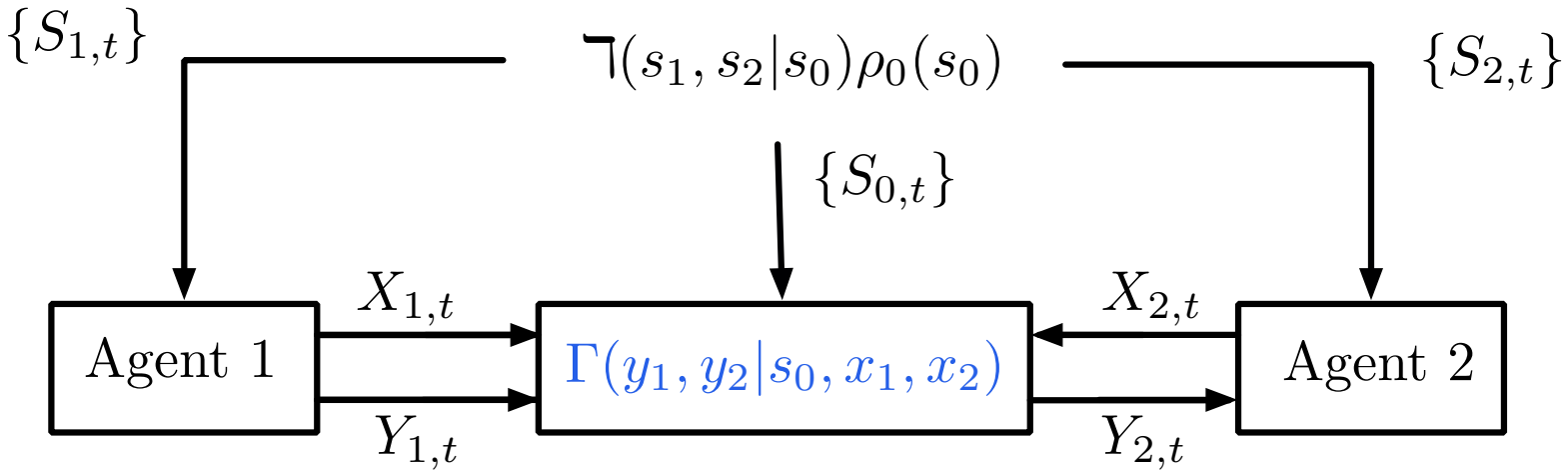
joint work with Benjamin Larousse and Samson Lasaulce

ITW 2015, Jerusalem, 27 April 2015

Coordination of Two Agents



Game-Theoretic Motivation

- Infinitely repeated game with 2 players 1 and 2
- Each player has observations about random nature state and other agents' actions
- Payoff-functions depend on both players' actions and nature state:

$$\omega_k(s_{0,t}, x_{1,t}, x_{2,t}), \qquad k \in \{1, 2\}$$

- Average expected payoff:

$$\begin{aligned}\overline{\omega}_k &\triangleq \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \mathbb{E}\left[\omega_k(S_{0,t}, X_{1,t}, X_{2,t})\right] \\ &= \sum_{(s_0,x_1,x_2)} \omega_k(s_0, x_1, \ldots x_2) \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} P_{X_{0,t}X_{1,t},X_{2,t}}(x_0, x_1, x_2).\end{aligned}$$

Setup

- state sequence $\{S_{0,t}\}$ i.i.d. $\sim \rho_0$

- Discrete-memoryless side-info: given $S_{0,t} = s_0$,

$$(S_{1,t}, \ldots, S_{K,t}) \sim \daleth(s_1, \ldots, s_K | s_0)$$

- Discrete memoryless network: $S_{0,t} = s_0, X_{1,t} = x_1, \ldots, X_{K,t} = x_K$,

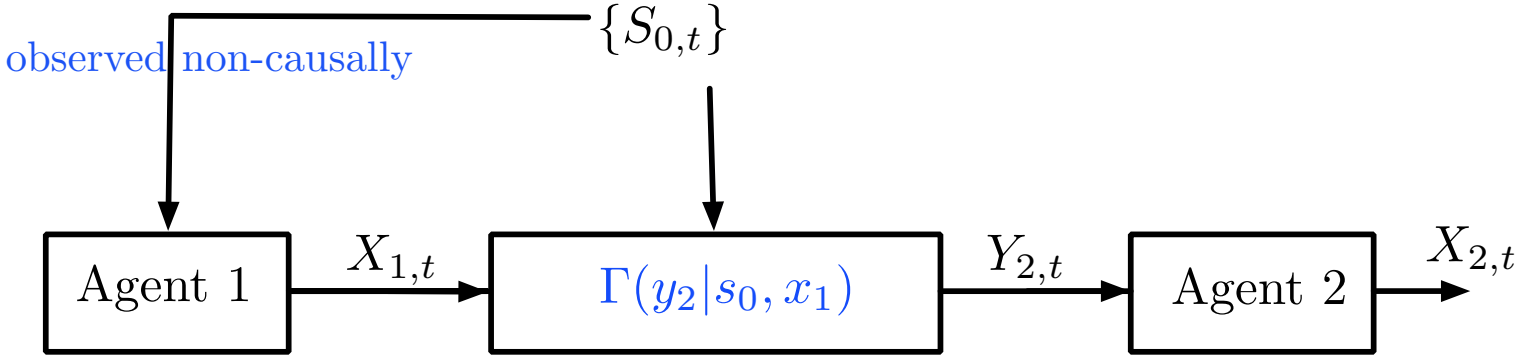
$$(Y_{1,t}, \ldots, Y_{K,t}) \sim \Gamma(y_1, \ldots, y_K | s_0, x_1, \ldots, x_K)$$

Coding and Implementable Distributions

- Causal or non-causal state-information:
  - Causal state-information: $X_{k,t} = \sigma_{k,t}^{(\mathsf{c})}(S_k^t, Y_k^{t-1})$
  - Non-causal state-information: $X_{k,t} = \sigma_{k,t}^{(\mathsf{nc})}(S_k^T, Y_k^{t-1})$
- Implementable distributions $\overline{Q}$: $\forall \epsilon > 0$ there exist T and encodings, s.t.

$$\left| \frac{1}{T} \sum_{t=1}^{T} P_{S_{0,t} X_{1,t} \cdots X_{K,t}}(s_0, x_1, x_2, \ldots, x_K) - \overline{Q}(s_0, x_1, x_2, \ldots, x_K) \right| \leq \epsilon.$$

Related and Previous Results, $K = 2$ Agents



- [Gossner et al. 2006, see also Zhao & Cuff 2011] solved:

  Agent 1 has S_0^T, Agent 2 observes $Y_{2,t} = (X_{1,t}, S_{0,t})$

- [Larousse, Lasaulce, Bloch 2013; related work by Le Treust 2014]:
  - Agent 1 has S_0^T, Agent 2 observes $Y_{2,t} = (X_{1,t}, S_{0,t})$
  - achievability for general $Y_{2,t}$

- [Cuff & Schieler 2011]: Agent 2 observes Y_2^T & channel state-independ.

- [Cuff & Permuter & Cover 2010]: empirical coordination (src coding)

Causal State-Info at all Agents

- Causal state-information: $X_{k,t} = \sigma^{(\mathsf{c})}_{k,t}(S^{t}_{k}, Y^{t-1}_{k})$

Theorem

$\overline{Q}$ implementable iff it factorizes as

$$\overline{Q}(s_0, x_1, x_2, \ldots, x_K)$$
$$= \sum_{u, s_1, s_2, \ldots, s_K} \left[\rho_0(s_0) \daleth(s_1, s_2, \ldots, s_K | s_0) P_U(u) \times \prod_{k=1}^{K} P_{X_k|US_k}(x_k | u, s_k) \right]$$

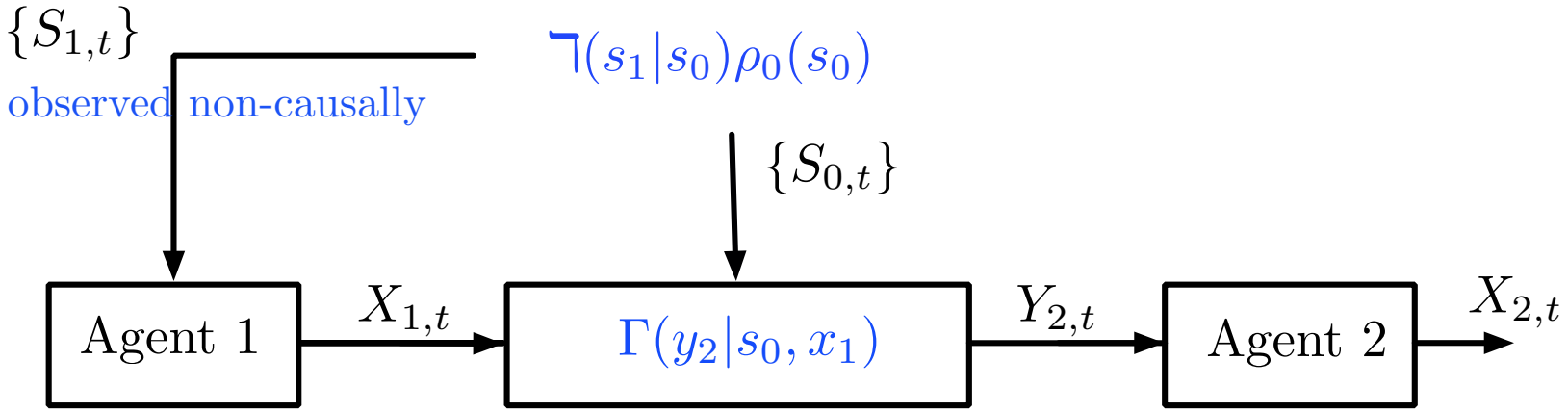
- No coding required:
  - Agree on a typical $\mathbf{u}$ sequence
  - Generate symbol-by-symbol $X_{k,t} \sim P_{X_k|US_k}(\cdot | u_t, s_{k,t})$

Non-Causal State-Info

- Non-causal state-information: $X_{k,t} = \sigma_{k,t}^{(\mathsf{c})}(S_k^T, Y_k^{t-1})$
- Includes general source-channel coding problem as special case
- Source-channel coding problem with transmitters $1, \ldots, \ell$ and receivers $\ell + 1, \ldots K$:
  - $S_0 = (S_1, \ldots, S_\ell)$
  - $S_{\ell+1} = \ldots = S_K = \emptyset$
  - $\Gamma(y_{\ell+1}, \ldots, y_K | x_1, \ldots, x_\ell)$

$\rightarrow$ in this case, strictly causal decoding constraint not limiting

Non-Causal State-Info and Point-to-Point Channel



Theorem

$\overline{Q}(s_0, x_1, x_2)$ *implementable iff it is marginal of some*

$$\begin{aligned} &Q(s_0, s_1, v, x_1, x_2, y_2) \\ &\quad = \rho_0(s_0)\daleth(s_1|s_0)P_{VX_1X_2|S_1}(v, x_1, x_2|s_1)\Gamma(y_2|s_0, x_1) \end{aligned}$$

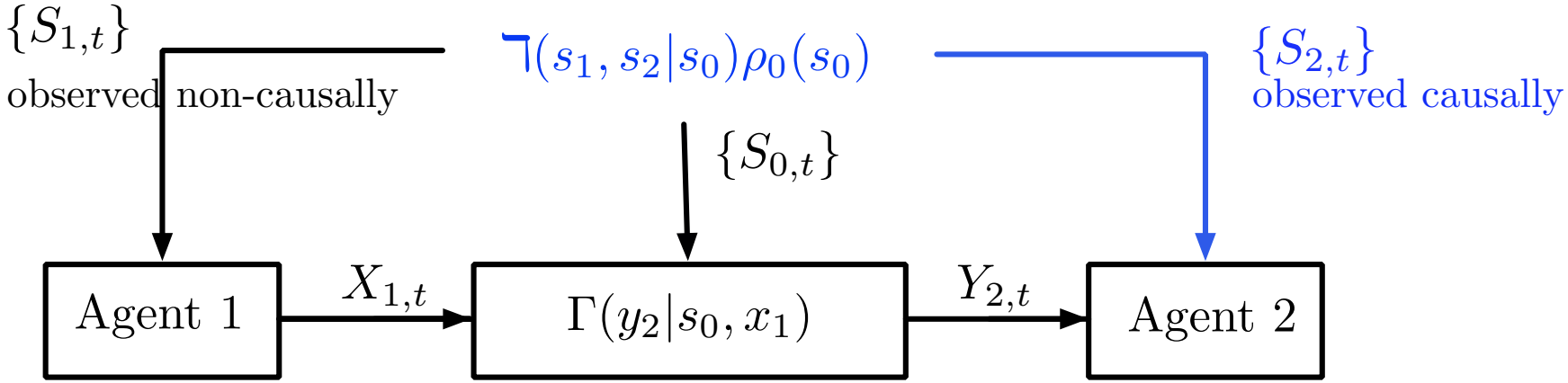
satisfying

$$I(S_1; X_2) \leq I(V; Y_2|X_2) - I(V; S_1|X_2)$$

Coding Scheme

- Blocks $b = 1, \ldots, B$
- Agent 1:
  - Creates Agent 2's action $\mathbf{x}_2^{(b)}$ coordinated with $\mathbf{s}_1^{(b)} \rightarrow$ covering index $J^{(b)}$
  - Creates $\mathbf{x}_1^{(b)}$ to be coordinated with $(\mathbf{x}_2^{(b)}, \mathbf{s}_1^{(b)})$ and to transmit $J^{(b+1)}$ $\rightarrow$ Gel'fand Pinsker code can do both!
- Agent 2:
  - Decode $J^{(b+1)}$ based on $(\mathbf{y}_2^{(b)}, \mathbf{x}_2^{(b)})$
  - Produce next block actions $\mathbf{x}_2^{(b+1)}(J^{(b+1)})$
- Successful if $I(S_1; X_2) \leq I(V; Y_2, X_2) - I(V; S_1, X_2)$

$K = 2$: Mixed Causal/Non-Causal State-Info



Theorem

$\overline{Q}(s_0, x_1, x_2)$ *implementable iff it is marginal of some*

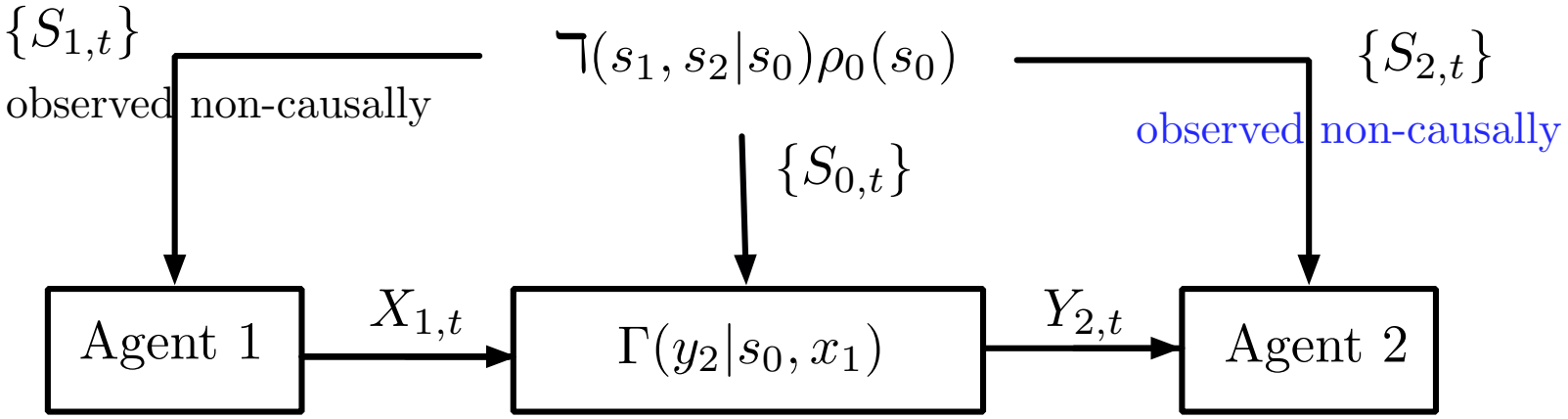
$$Q(s_0, s_1, s_2, u, v, x_1, x_2, y_2)$$
$$= \rho_0(s_0)\daleth(s_1, s_2|s_0)P_{UVX_1|S_1}(u, v, x_1|s_1)P_{X_2|US_2}(x_2|u, s_2)\Gamma(y_2|s_0, x_1)$$

satisfying

$$I(S_1; U) \leq I(V; Y_2, S_2|U) - I(V; S_1|U)$$

- aux. $\mathbf{u}^{(b)}$ (instead of $\mathbf{x}_2^{(b)}$) coordinated with $\mathbf{s}_1^{(b)}$ and described to Ag. 2

$K = 2$: All Non-Causal State-Info



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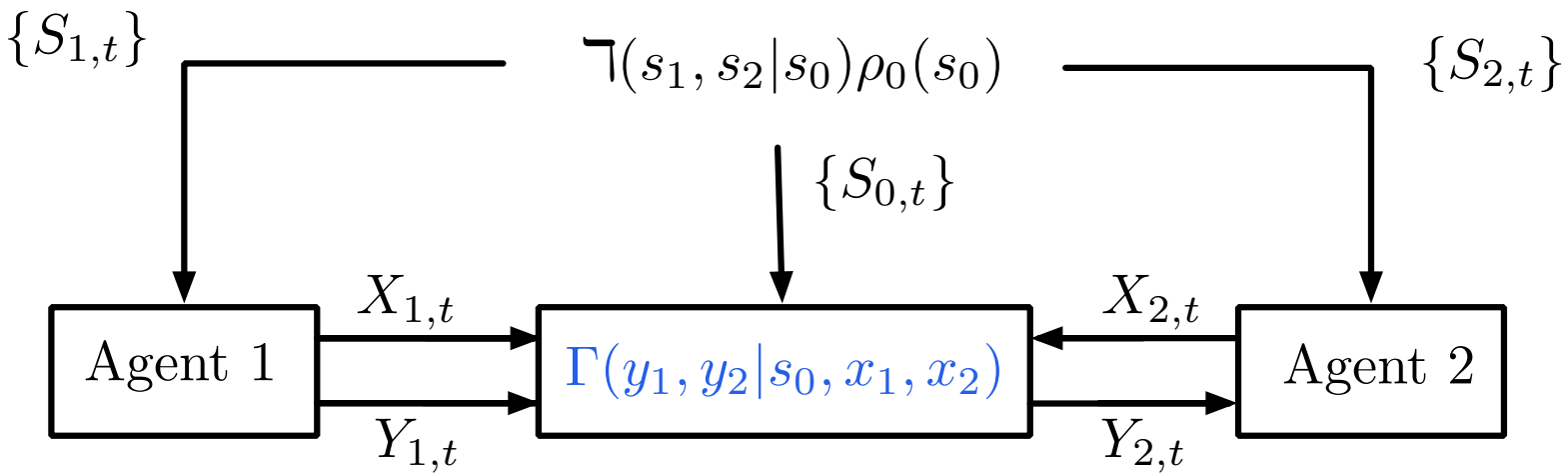
satisfying

$$I(S_1; U|S_2) \leq I(V; Y_2, S_2|U) - I(V; S_1|U)$$

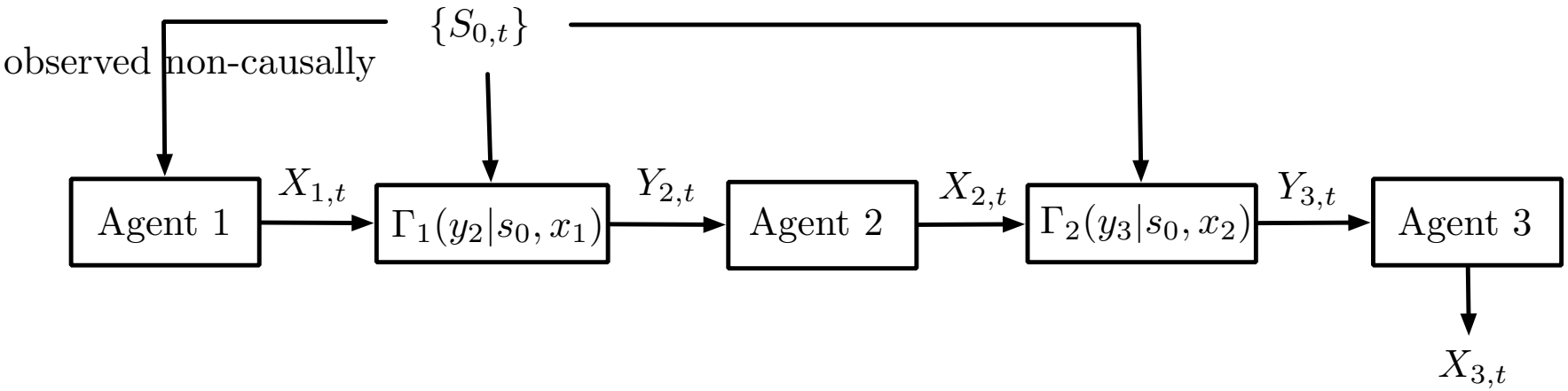
- Wyner-Ziv codeword $\mathbf{u}^{(b)}$

Two-Way Channel $\Gamma(y_1, y_2|s_0, x_1, x_2)$



- Result with causal state-info at Agent 2 remains valid
- Result with non-causal state-info at Agent 2 remains valid if $S_2 = f(S_1)$ $\rightarrow$ all relevant communication from Agent 1 to 2
- New scheme required with symmetric operations and two-way source and channel codes

Cascade Network with Non-Causal Side-Info



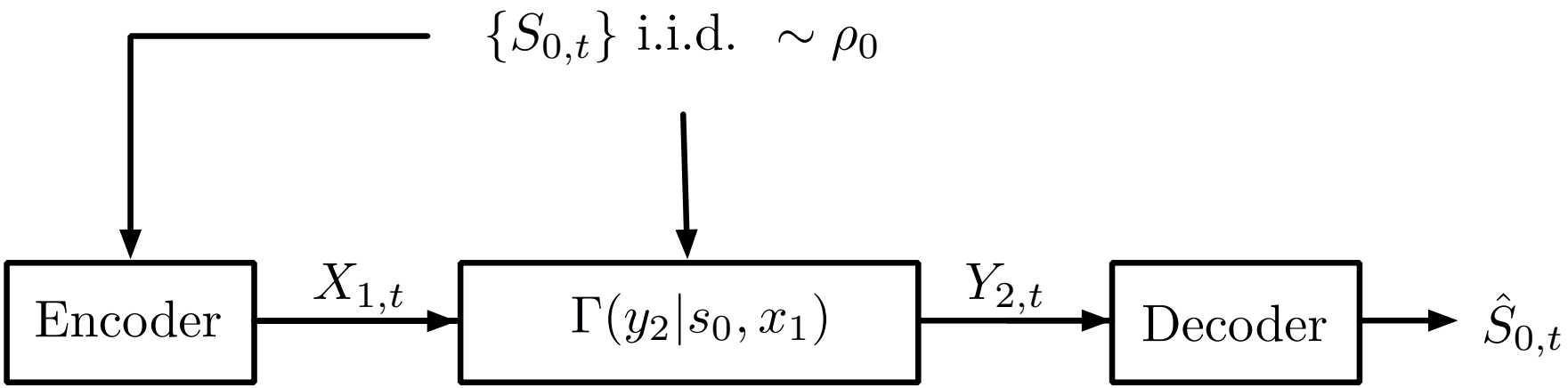
- $\overline{Q}(s_0, x_1, x_2, x_3)$ implementable if $\exists P_{V_1|S_0X_1X_2X_3}$ s.t.

$$I(S_0; X_2, X_3) \leq I(V_1; Y_2|X_2, X_3) - I(V_1; S_0|X_2, X_3)$$
$$I(S_0; X_3) \leq I(X_2; Y_3|X_3) - I(X_2; S_0|X_3)$$

Coding Scheme for Cascade Network

- Agent 1:
  - Coordinate $\left(\mathbf{s}_0^{(b)}, \mathbf{x}_3^{(b)}(J^{(b)}), \mathbf{x}_2^{(b)}(J^{(b+1)}, K^{(b)}|J^{(b)})\right)$
  - Use GP-code to coordinate also $\mathbf{x}_1^{(b)}$ and send $(J_{b+2}, K^{(b+1)})$ to Agent 2
- Agent 2: Send $\mathbf{x}_2^{(b)}(J^{(b+1)}, K^{(b)}|J^{(b)})$ and decode $(J^{(b+2)}, K^{(b+1)})$
- Agent 3: Send $\mathbf{x}_3^{(b)}(J^{(b)})$ and decode $J^{(b+1)}$

State-Communication and Relation to Coordination



- Distortion D achievable if $\overline{\lim}_{T\to\infty} \left(\frac{1}{T} \sum_{t=1}^{T} \mathsf{E}\left[d(S_{0,t}, \hat{S}_{0,t}) \right] \right) \leq D$

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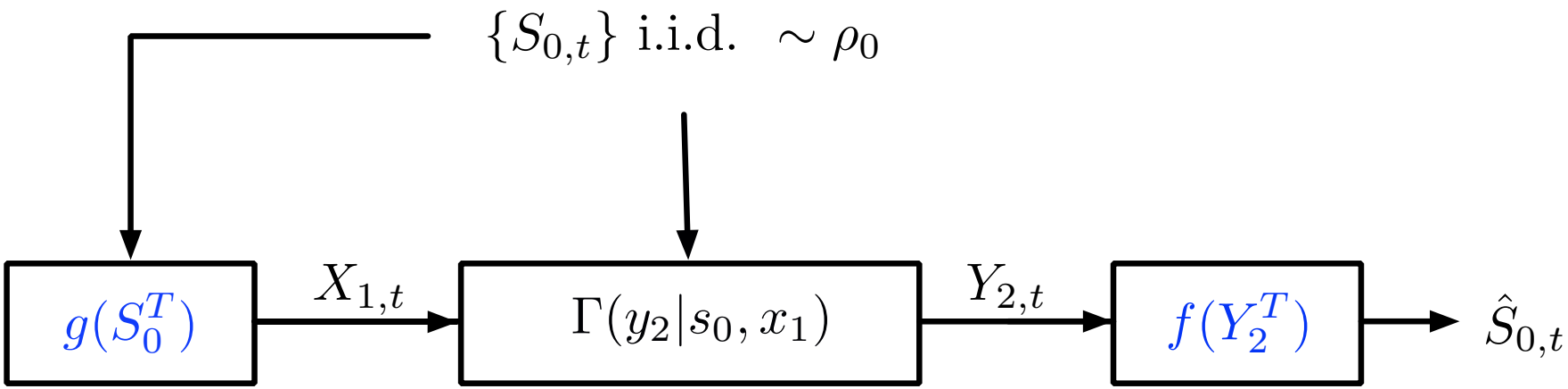

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- Non-causal encoding and decoding: achievability [Sutivong et al. 2005]

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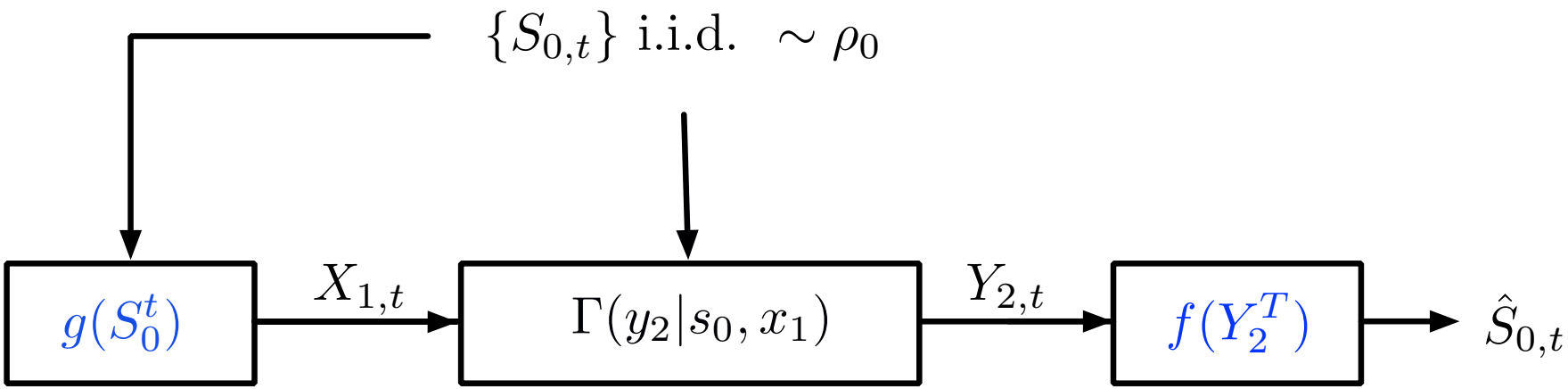

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- Causal or strictly causal encoding/non-causal decoding: [Choudhuri et al. 2013]

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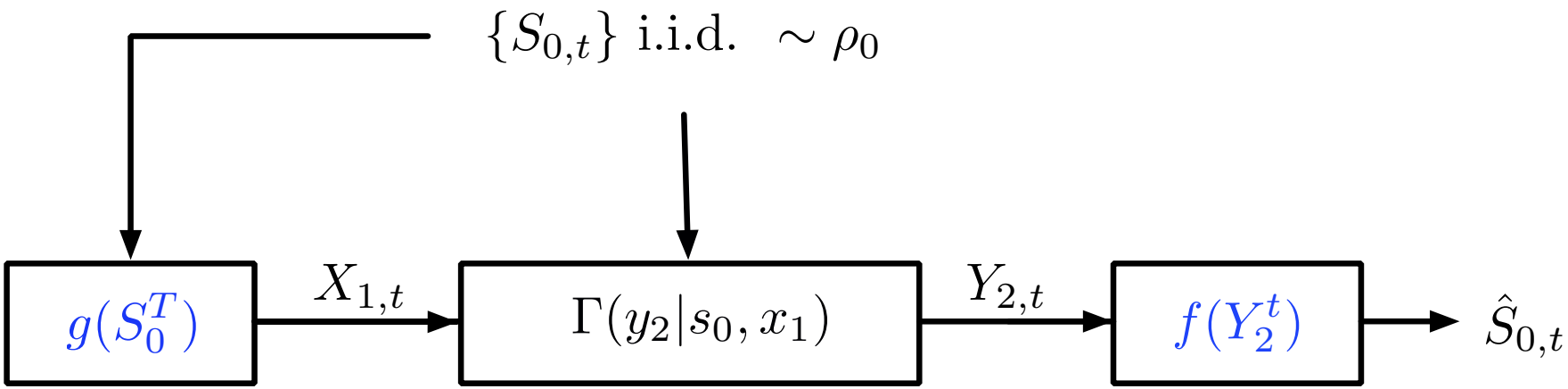

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- Non-causal encoding / causal or strictly causal decoding

State-Communication with (Strictly) Causal Decoding

- Results follow directly from our coordination results

- Non-causal encoding, causal decoding: D achievable iff,

$$\mathsf{E}[S_0, g(U, Y_2)] \leq D$$

for $(S_0, U, V, X_1, Y_2) \sim \rho_0(s_0) P_{UVX_1|S_0}(u, v, x_1|s_0)\Gamma(y_2|s_0, x_1)$ s.t.

$$I(U; S_0) \leq I(V; Y_2|U) - I(V; S_0|U)$$

State-Communication with (Strictly) Causal Decoding

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$$I(U; S_0) \leq I(V; Y_2|U) - I(V; S_0|U)$$

- Non-causal encoding, causal decoding: result as above but with $g(U)$

Comparison of Various Causality Constraints

	non-causal	causal
encoding	GP coding	Shannon strategies
decoding	WZ coding	standard source coding

Mutual information constraints:

- Causal encoding/non-causal decoding:

  $I(U;S_0|V) - I(U;Y_2|V) \leq I(V;Y_2)$ for $p_V p_{S_0} p_{U|VS_0} p_{X_1|VS_0}$

- Non-causal encoding/non-causal decoding: (achievable)

  $I(U;S_0|V) - I(U;Y_2|V) \leq I(V;Y_2) - I(V;S_0)$ for $p_{UVS_0X_1}$

Summary/Outlook

- Coordination of agents over state-dependent networks
- Set of implementable distributions when all agents have causal state-information
- Set of implementable distributions for some 2-agents setup
- Subset of implementable distributions over a 3-agents cascade network
- State-communication with causal and strictly-causal decoding